

Computer Vision-Based Descriptive Analytics of Seniors' Daily Activities for Long-Term Health Monitoring

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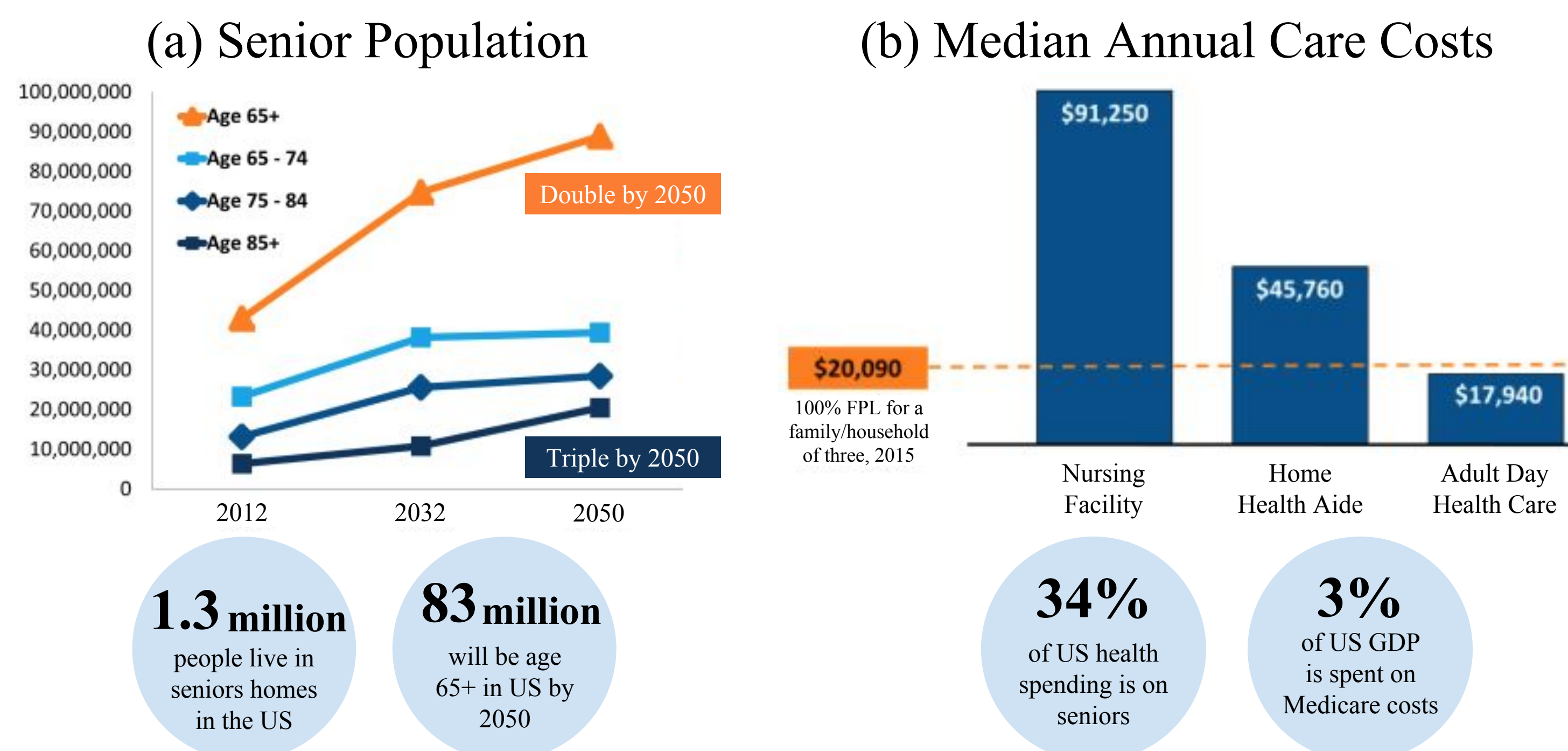
¹ Stanford University, ² On Lok Inc.

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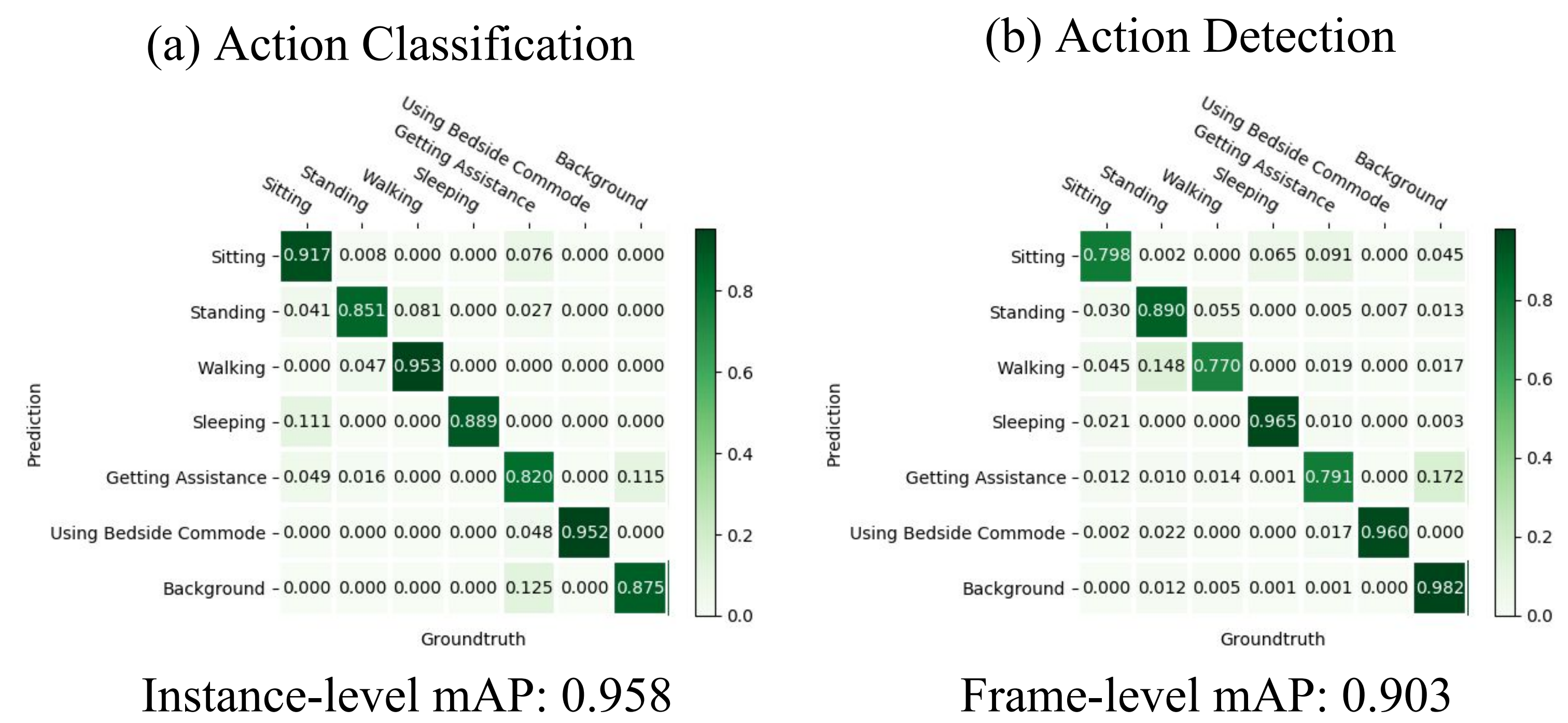
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Background

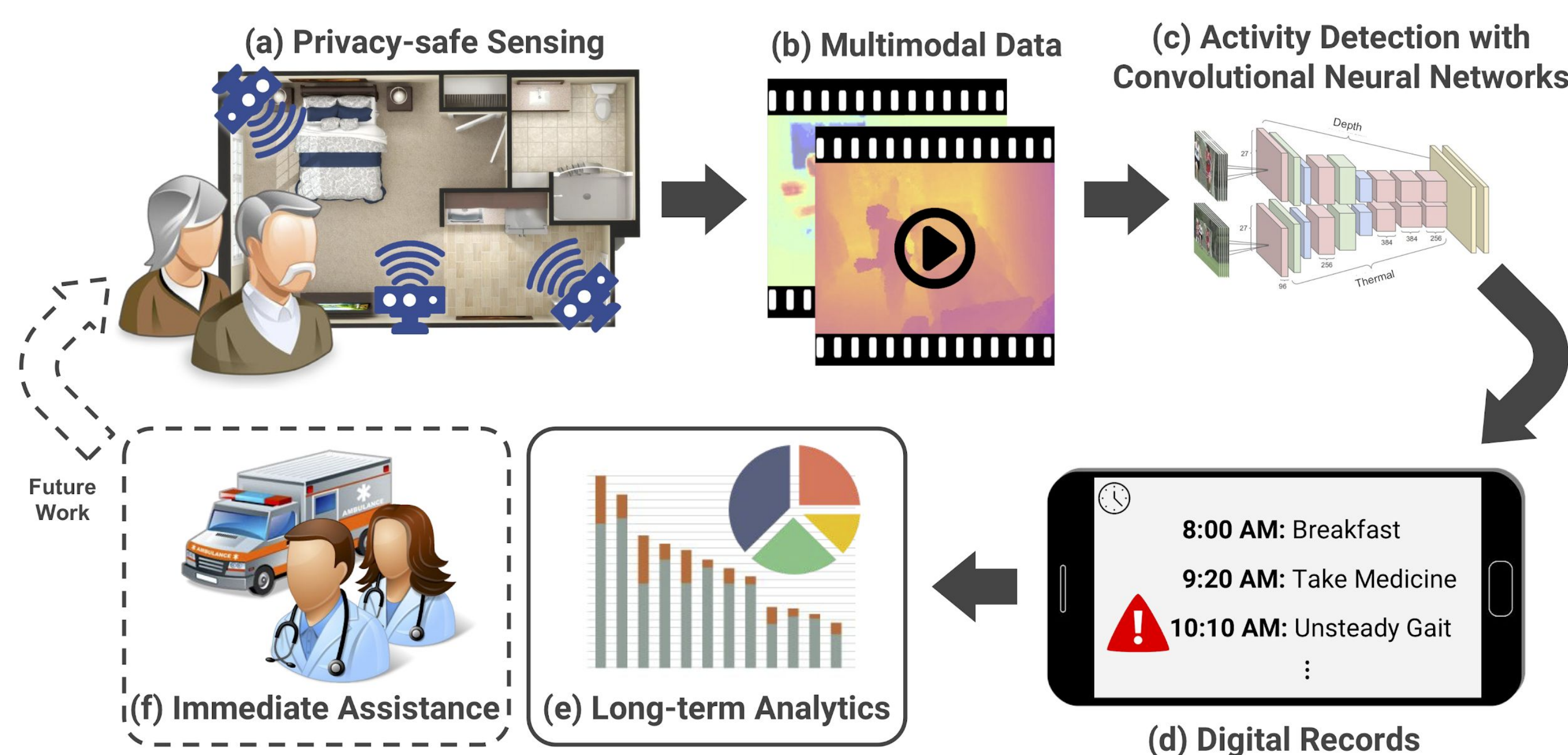


Quantitative Result



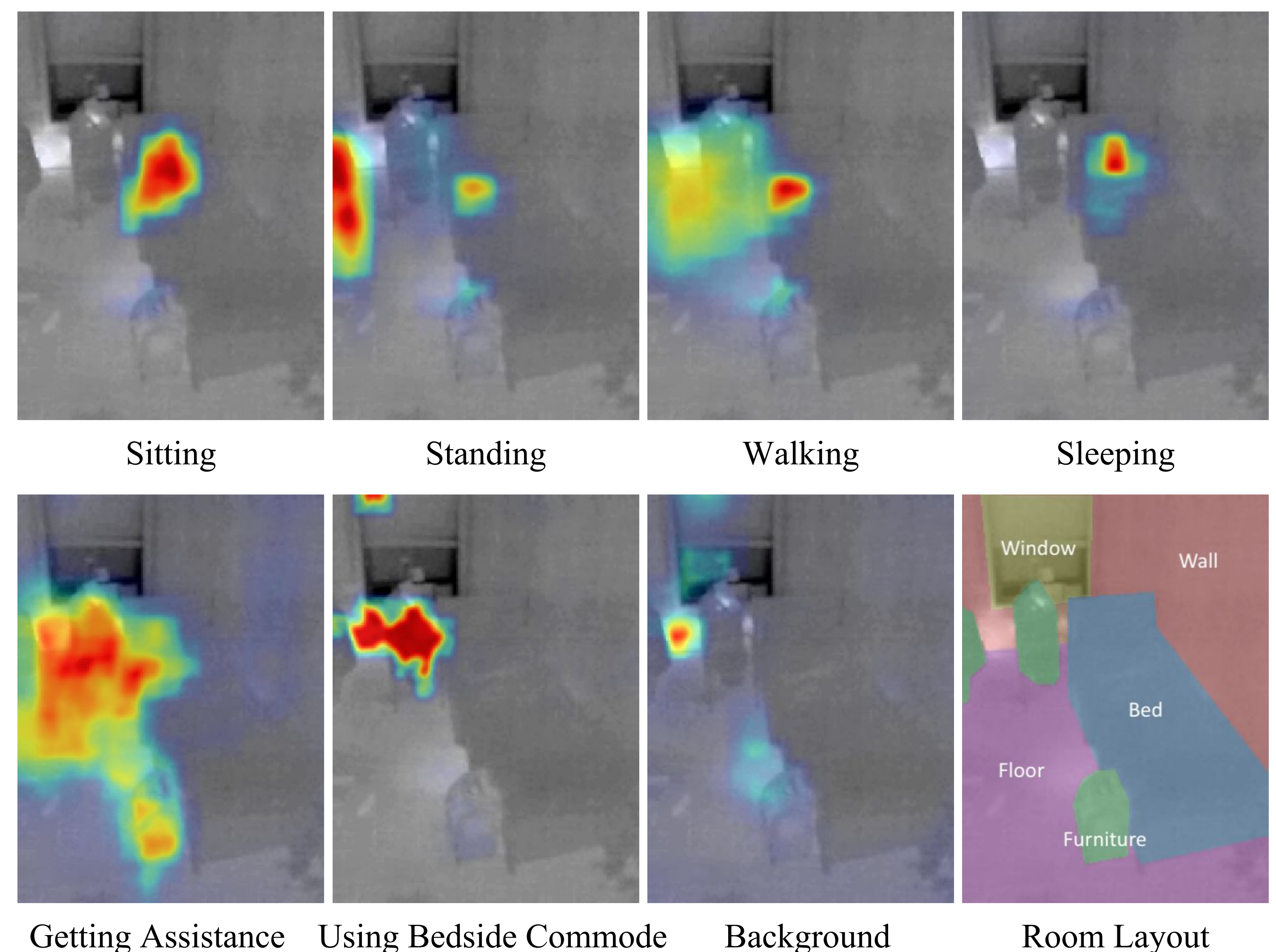
Objective

Our goal is to build a computer vision-based approach that leverages non-intrusive, privacy-compliant, multimodal sensors to continuously detect seniors' activities and provide the corresponding long-term descriptive analytics.

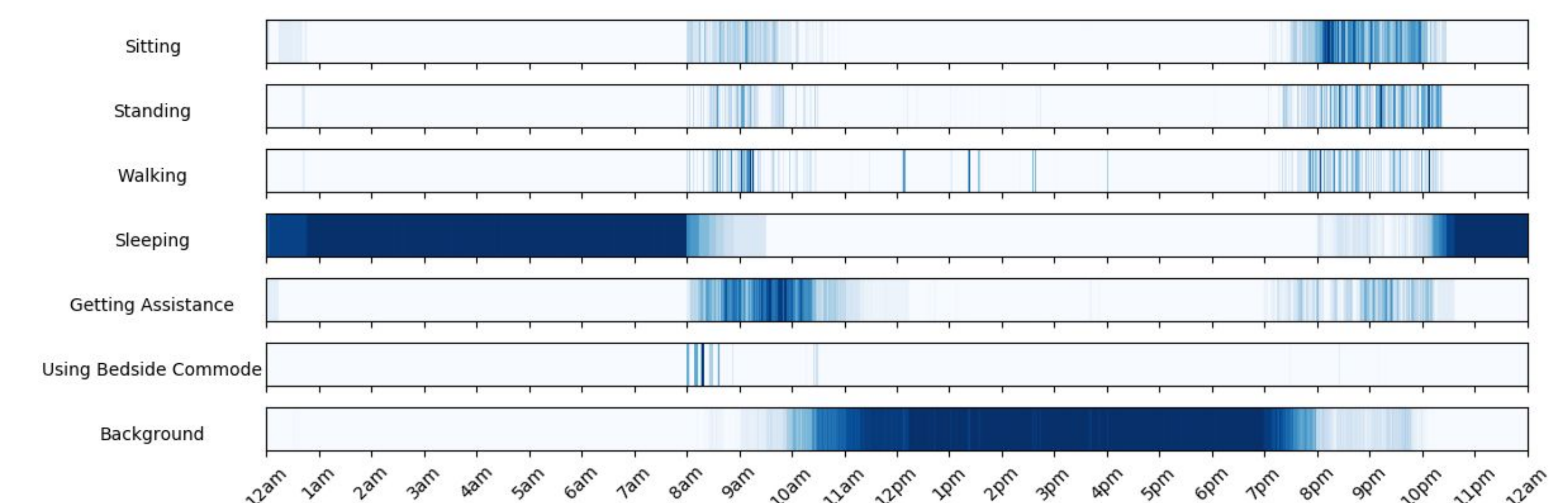


Descriptive Analytics

(a) Spatial Heatmaps

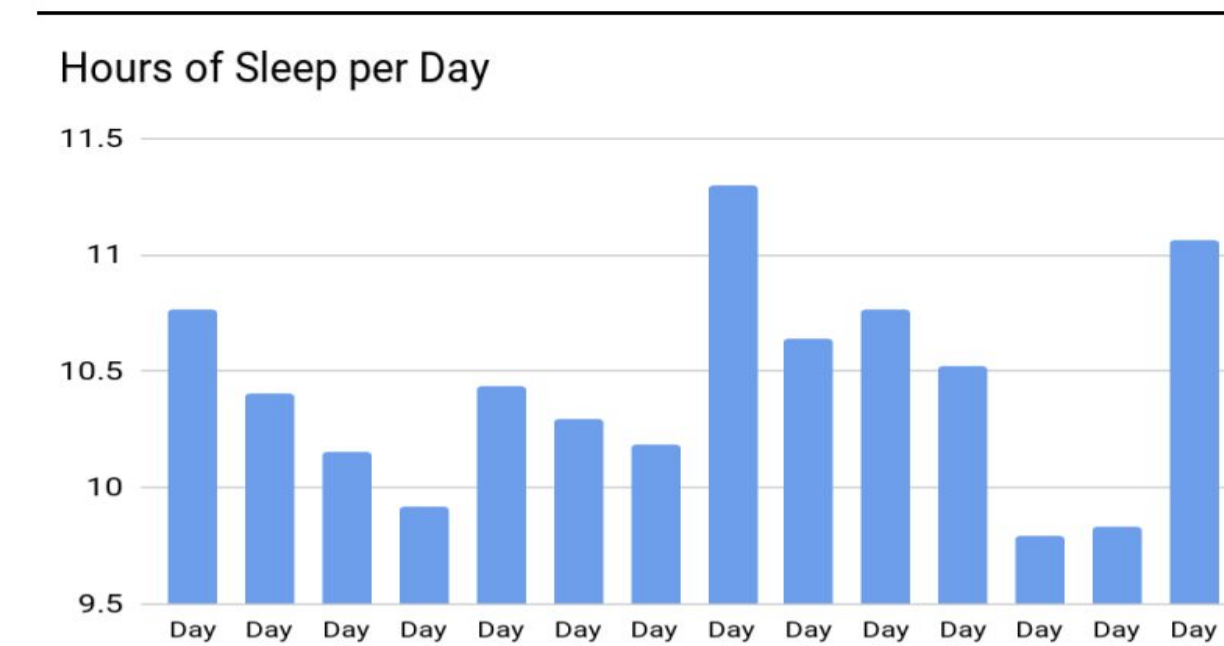


(b) Temporal Heatmaps

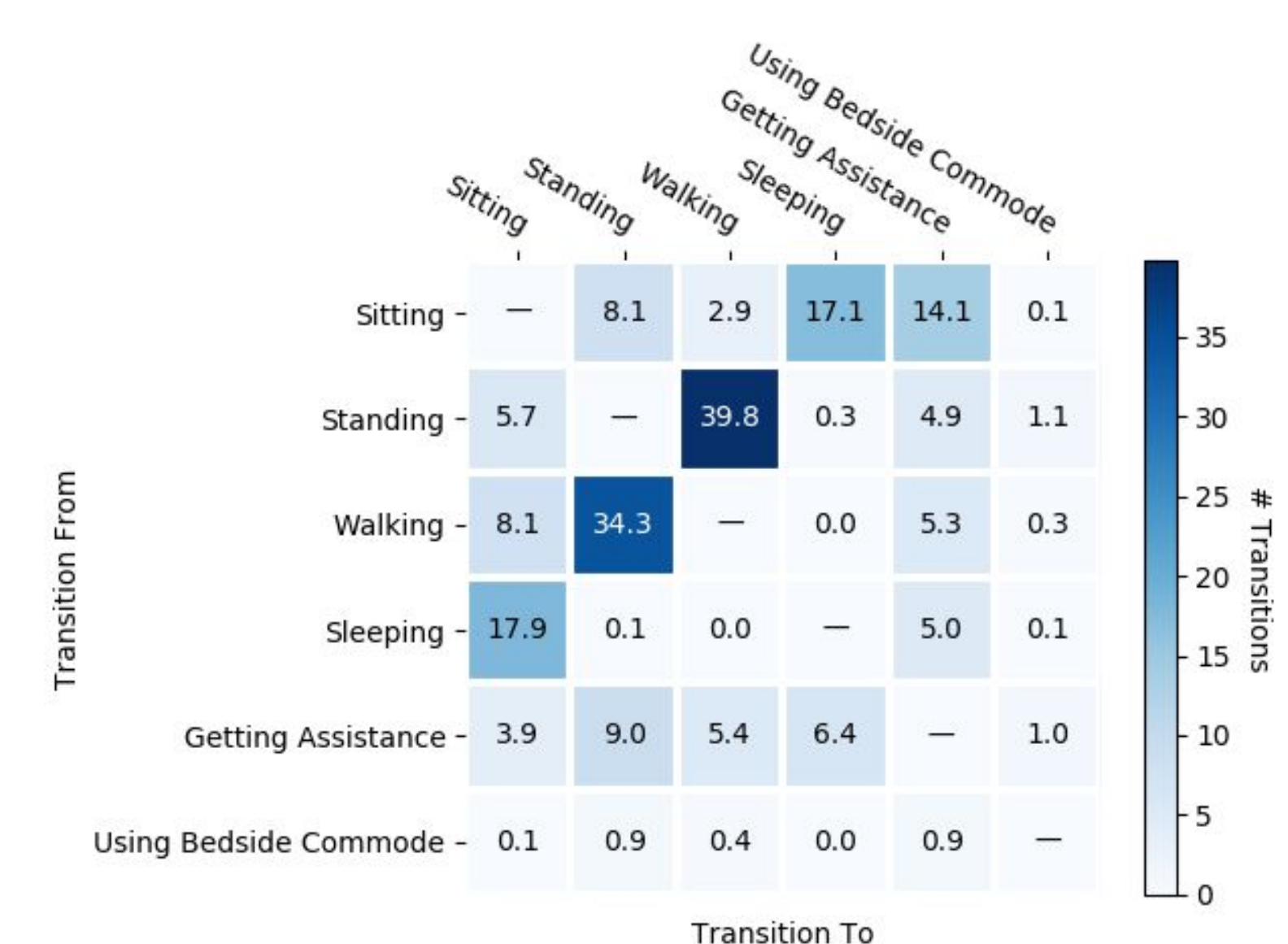


(c) Duration and # of Instances

Activity	Duration (min)	# instances
Sitting	81.75 ± 30.06	82.36 ± 67.28
Standing	27.23 ± 7.84	70.71 ± 16.52
Walking	4.76 ± 1.01	54.50 ± 11.10
Sleeping	626.11 ± 25.90	28.86 ± 18.13
Getting Assistance	114.01 ± 34.98	68.93 ± 31.05
Using Bedside Commode	1.69 ± 1.99	2.86 ± 4.76

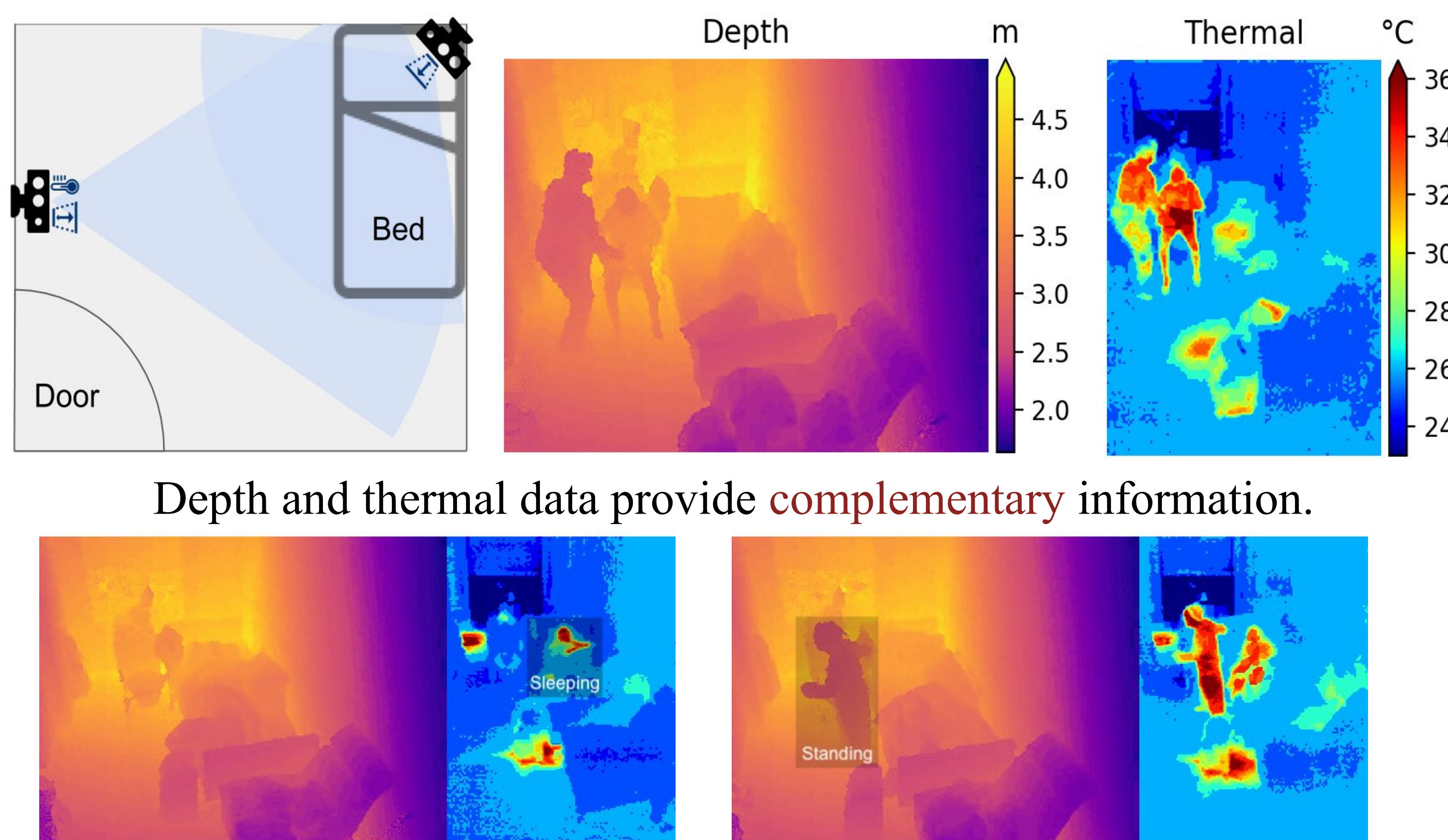


(d) Transitions

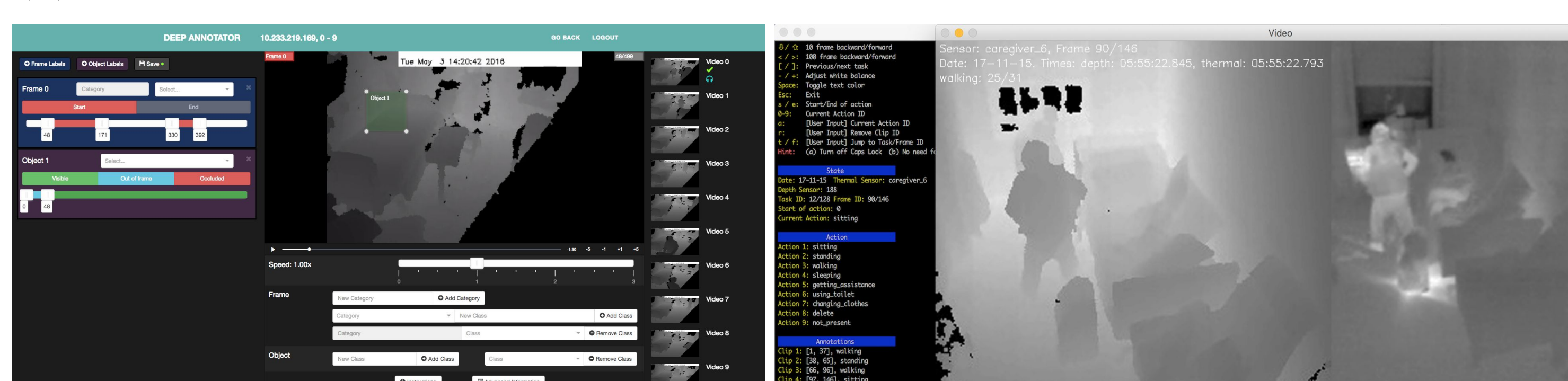


Method

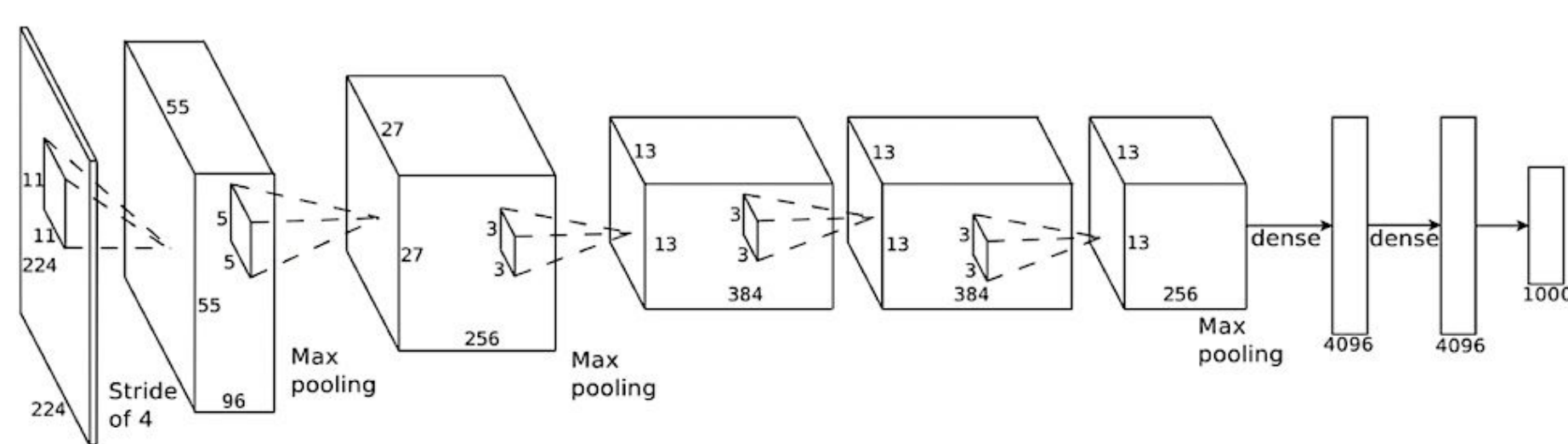
(a) Long-term video data collection via privacy-safe, multimodal sensors.



(b) Combination of automated and manual data annotation.



(c) Action classification and detection with deep learning models.



Conclusion

The system is helpful in objectively recording and analyzing long-term behaviors and capturing seniors' health decline. Our work is progress towards a smart senior home that uses computer vision to support caregivers in senior healthcare to help meet the challenges of an aging worldwide population.